EDUCATION PROJECT REPORT

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1. Introduction

Educational opportunity is not spread evenly across U.S. communities. Household income, employment, education levels, and family structure can all be linked to how well students do in school. This project asks how well socioeconomic factors predict average ACT scores at U.S. high schools, and whether knowing a school's charter status adds anything once those factors are accounted for.

I start by exploring the data, then compare two multiple linear regression models on the same held-out test set. Model 1 uses five socioeconomic predictors; Model 2 adds an indicator for charter status. The goal is prediction and description, not estimating causal effects.

2. Data and Preparation

2.1 Data sources

The first file, EdGap_data.xlsx, has 7,986 observations and seven fields on school ACT performance and socioeconomic conditions. EdGap pulls its socioeconomic indicators from the U.S. Census Bureau's American Community Survey and its test results from public state education sources. The second file, ccd_sch_029_1617_w_la_11212017.csv, comes from the National Center for Education Statistics (NCES) Common Core of Data for 2016–2017. It has 102,183 observations and 65 fields and supplies school identifiers and characteristics, including charter status. EdGap warns that testing requirements differ by state, which limits direct comparison of average scores across states.

2.2 Data cleaning and merging

In education_data_preparation.ipynb, I loaded the Excel and CSV files with pandas, checked dimensions, data types, and missing values, and kept the relevant school and socioeconomic fields. I converted the school identifiers to a common format and merged the two datasets with a left join. I then selected high schools, checked for duplicate records and implausible values, set invalid ACT scores and lunch percentages to missing, and dropped records with no valid outcome. Missing numerical predictors were filled in with scikit-learn's IterativeImputer. Finally, I renamed the columns to lowercase snake_case and exported the tidy, school-level table to data/education_clean.csv.

The cleaned file has 7,227 school records and 15 columns. When loaded with pandas, 6,858 records are labeled "No" for charter status, 170 are labeled "Yes," and 199 are missing or unspecified. For the regression comparison I excluded rows with missing charter status, so both models use the same complete-case subset. The modeling notebook also created income_10k (median_income divided by 10,000) and lunch_pct_points (percent_lunch times 100). These rescale the predictors without changing the information they carry.

3. Exploratory Data Analysis

In education_eda.ipynb, I used pandas, Seaborn, and Matplotlib to look at summary statistics, a numerical correlation matrix, pair plots, and boxplots. I used these to judge the direction and rough shape of relationships, spot possible outliers, and check whether the predictors were correlated with one another.

The strongest relationship was between school lunch participation and average ACT score, with Pearson's $r \approx -0.785$. Schools with higher free or reduced-price lunch participation tended to have lower average ACT scores. That is a strong negative association in this dataset, but it does not show that lunch participation causes lower scores. The socioeconomic predictors also overlap conceptually, so their individual regression coefficients should not be read as independent causal effects.

4. Regression Methods

The response variable was average_act. The baseline model used rate_unemployment, percent_college, percent_married, income_10k, and lunch_pct_points. The expanded model added charter_yes, coded 1 when charter = "Yes" and 0 when charter = "No." The 199 unspecified records were dropped from both models by the complete-case filter; I did not treat them as confirmed non-charter schools.

Both models were fit with scikit-learn's LinearRegression. I split the complete-case data 80/20 into training and test sets using train_test_split(test_size=0.2, random_state=42), fit each model on the training data, and predicted on the same test observations. I evaluated them with mean absolute error (MAE), root mean squared error (RMSE), and R-squared (R^2). Lower MAE and RMSE mean smaller prediction errors, and a higher R^2 means the model does better relative to simply predicting the mean.

The baseline model has the form $ACT_i = \beta_0 + \beta_1$ unemployment_i + β_2 college_i + β_3 married_i + β_4 income_i + β_5 lunch_i + ε_i. The expanded model adds a term, β_6 charter_yes_i. I chose charter status because it is a school characteristic from NCES, separate from the EdGap socioeconomic measures, and it might carry extra information about school-level ACT performance.

5. Results

Test-set model	MAE	RMSE	R^2
Socioeconomic factors	1.161	1.554	0.615
Socioeconomic factors + charter	1.161	1.554	0.615

Table 1. Model evaluation metrics reported by the Jupyter notebook, rounded to three decimal places.

The socioeconomic model had an MAE of 1.161 ACT points and an RMSE of 1.554 points. Its test R^2 of 0.615 means the five socioeconomic variables carry a substantial amount of predictive information, though a good deal of variation is still unexplained. The actual-versus-predicted scatterplot shows a positive trend, but predictions are generally less extreme than the observed scores: the model tends to overpredict at the low end and underpredict at the high end.

Adding charter_yes left MAE, RMSE, and R^2 unchanged at three decimal places, so charter status gave no detectable improvement in test performance on this split. That does not mean charter status is unrelated to ACT scores. Only 170 of the 7,227 cleaned records are charter schools, and the evaluation rests on a single random split.

6. Limitations and Conclusion

Several limitations affect how these results should be read. First, EdGap notes that states differ in who takes the ACT or SAT, which makes school averages hard to compare. Second, the data are observational and school-level, so the relationships say nothing definitive about individual students or about causes. Third, I imputed missing numerical values during data preparation, before the train/test split. Because imputation can draw on information from rows that later end up in the test set, this can leak information; a stricter evaluation would fit the imputer on training data only. Fourth, a single split does not show how much performance varies across samples. Finally, charter schools are rare in this dataset, which limits the comparison.

Overall, the results support the conclusion that the selected socioeconomic factors help predict average ACT scores in this dataset: the baseline regression reached a test R^2 of 0.615 and an MAE of 1.161. The strong negative correlation between lunch participation and ACT scores fits that picture. Adding charter status completed the project's additional-predictor analysis but did not materially improve the reported metrics. Future work could try other school-level predictors and use cross-validation with training-only imputation to get more reliable estimates of predictive performance.

References

EdGap. (n.d.). About the EdGap map and data. https://www.edgap.org/

National Center for Education Statistics. (n.d.). Common Core of Data (CCD). https://nces.ed.gov/ccd/